Bounded symplectic embeddings extend to symplectic automorphisms, and commuting pairs admit commuting extensions

Abstract

Let $H = \ell^2(\mathbb{N}; \mathbb{R}^{2d}) = \bigoplus_{n=1}^{\infty} \mathbb{R}^{2d}$ carry the strong symplectic form $\Omega = \bigoplus_n \omega_0$ associated with the orthogonal complex structure $\mathcal{J} = \bigoplus_n J_{2d}$. We prove that every bounded linear map $T : H \to H$ satisfying $T^*\mathcal{J}T = \mathcal{J}$ (a symplectic embedding) extends to a bounded symplectic automorphism of a larger symplectic Hilbert space. We then prove that if T and T' are commuting symplectic embeddings, they admit commuting extensions to symplectic automorphisms of a common enlargement. The single-operator result is elementary and constructive. The commuting-pair result follows an Itô-type conjugation argument on that model, with boundedness obtained from the Uniform Boundedness Principle and the structure of the symplectic Wold decomposition.

Contents

1 Introduction and statement of results

Let V be a real separable Hilbert space with inner product g and norm $\|\cdot\|$. An orthogonal complex structure on V is a bounded operator $\mathcal{J}$ satisfying

$$\mathcal{J}^2 = -I, \qquad \mathcal{J}^* = -\mathcal{J}. \tag{1}$$

The associated bilinear form

$$\Omega(x, y) := g(\mathcal{J}x, y) \tag{2}$$

is then a *strong* symplectic form: the map $x \mapsto \Omega(x, \cdot)$ is a topological isomorphism of V onto V^*.

The model space of interest is

$$H = \ell^2(\mathbb{N}; \mathbb{R}^{2d}) = \bigoplus_{n=1}^{\infty} \mathbb{R}^{2d}, \tag{3}$$

with $\mathcal{J} = \bigoplus_n J_{2d}$ the direct sum of the standard complex structure on $\mathbb{R}^{2d}$. Then $\Omega = \bigoplus_n \omega_0$ is the direct sum of the standard area forms.

Definition 1.1. A bounded linear operator $T : V \to V$ is a *symplectic embedding* if

$$\Omega(Tx, Ty) = \Omega(x, y) \qquad \text{for all } x, y \in V, \tag{4}$$

or equivalently $T^*\mathcal{J}T = \mathcal{J}$. It is a *symplectic automorphism* if it is a symplectic embedding and is bijective (equivalently, surjective); in that case $T \in \mathrm{Sp}(V, \Omega)$ and T^{-1} is automatically bounded.

In finite dimensions every symplectic embedding is an automorphism. In infinite dimensions this fails: the unilateral block shift on H is a symplectic embedding that is not onto. The first theorem is the symplectic analogue of extending an isometry to a unitary.

Theorem 1.2 (Single embedding)**.** *Let $T : V \to V$ be a symplectic embedding on a separable real symplectic Hilbert space (V, Ω) as above. Then there exist a symplectic Hilbert space $(W, \widehat{\Omega})$ containing (V, Ω) as a closed symplectic subspace, and a symplectic automorphism $\widetilde{T} : W \to W$, such that $\widetilde{T}|_V = T$.*

The second theorem is the symplectic analogue of Itô's theorem on commuting isometries.

Theorem 1.3 (Commuting pair)**.** *Let $T, T' : V \to V$ be commuting symplectic embeddings. Then there exist a symplectic Hilbert space $(W, \widehat{\Omega})$ containing (V, Ω), and commuting symplectic automorphisms $\widetilde{T}, \widetilde{T}' : W \to W$, such that $\widetilde{T}|_V = T$ and $\widetilde{T}'|_V = T'$.*

Remark 1.4. The same conclusion holds for any finite (or countable) family of mutually commuting symplectic embeddings: the argument below extends verbatim to representations of the semigroup $\mathbb{N}^k$.

2 Preliminaries

Throughout, $(V, g, \mathcal{J}, \Omega)$ is as in the introduction.

Lemma 2.1 (Basic identities)**.** *(i) $\|\mathcal{J}x\| = \|x\|$ for every $x \in V$.*

(ii) T satisfies (4) if and only if $T^*\mathcal{J}T = \mathcal{J}$.

(iii) Ω is strong: $\flat : V \to V^*$, $\flat(x) = \Omega(x, \cdot)$, is a topological isomorphism.

Proof. (i) From (1), $\mathcal{J}^*\mathcal{J} = (-\mathcal{J})\mathcal{J} = -\mathcal{J}^2 = I$, so $\|\mathcal{J}x\|^2 = g(x, \mathcal{J}^*\mathcal{J}x) = \|x\|^2$.

(ii) $\Omega(Tx, Ty) = g(\mathcal{J}Tx, Ty) = g(T^*\mathcal{J}Tx, y)$ and $\Omega(x, y) = g(\mathcal{J}x, y)$. Equality for all y yields $T^*\mathcal{J}T = \mathcal{J}$.

(iii) $\flat = \iota_g \circ \mathcal{J}$ with ι_g the Riesz isomorphism, and both factors are topological isomorphisms. □

Lemma 2.2 (Left inverse and closed range)**.** *Let T be a symplectic embedding. Then:*

(i) T is bounded below: $\|Tx\| \geq \|x\|/\|T\|$ for all x.

(ii) T is injective and $R := \operatorname{ran} T$ is closed.

(iii) The operator $L := -\mathcal{J}T^*\mathcal{J}$ satisfies $LT = I$ and $\|L\| \leq \|T\|$.

(iv) TL is the bounded projection onto R along the symplectic complement defined in Lemma 3.1 below; writing $E := TL$ and $F := I - E$, one has $E|_R = \mathrm{id}_R$ and $\ker E = R^{\perp_\Omega}$.

Proof. (i) By Lemma 2.1,

$$\|x\| = \|\mathcal{J}x\| = \|T^*\mathcal{J}Tx\| \le \|T\|\,\|Tx\|.$$

(ii) Boundedness from below yields injectivity and closed range by the usual Cauchy sequence argument. (iii) $LT = (-\mathcal{J}T^*\mathcal{J})T = -\mathcal{J}(T^*\mathcal{J}T) = -\mathcal{J}\mathcal{J} = I$. Also $\|L\| \le \|\mathcal{J}\|\,\|T\|\,\|\mathcal{J}\| = \|T\|$. (iv) $E^2 = TLTL = T(LT)L = TL = E$, $\operatorname{ran} E = R$, and $Ev = 0 \Leftrightarrow Lv = 0 \Leftrightarrow T^*\mathcal{J}v = 0 \Leftrightarrow \mathcal{J}v \in \ker T^* \Leftrightarrow v \in \mathcal{J}^{-1}(R^{\perp_g}) = R^{\perp_\Omega}$. □

3 Symplectic complement of the range

Lemma 3.1 (Symplectic complement)**.** *Let T be a symplectic embedding, $R = \operatorname{ran} T$, and*

$$N := R^{\perp_\Omega} = \{v \in V : \Omega(v, r) = 0 \textit{ for all } r \in R\}.$$

Then:

(i) $N = \mathcal{J}^{-1}(R^{\perp_g})$ is closed.

(ii) $V = R \oplus N$ as a topological direct sum, and $\Omega(R, N) = 0$.

(iii) $(R, \Omega|_R)$ and $(N, \Omega|_N)$ are strong symplectic Hilbert spaces.

(iv) $T : V \to R$ is a symplectic isomorphism (bicontinuous).

(v) T is surjective if and only if $N = \{0\}$.

Proof. (i) Immediate from $\Omega(v, r) = g(\mathcal{J}v, r)$.

(ii) Let P_R be the g-orthogonal projection onto R and set $B := P_R\mathcal{J}|_R : R \to R$. For $r, s \in R$,

$$g(Br, s) = g(\mathcal{J}r, s) = \Omega(r, s),$$

so $B = \iota_R^{-1} \circ \flat_R$. By (iii) below $\flat_R$ is an isomorphism, hence so is B. The operator $E := B^{-1}P_R\mathcal{J}$ is a bounded idempotent with range R and kernel N (same computation as in Lemma 2.2(iv) with this formula for E). Thus $V = R \oplus N$ topologically and $\Omega(R, N) = 0$ by definition of N. (The projection E coincides with TL from Lemma 2.2.)

(iii) Nondegeneracy of $\Omega|_N$: if $n \in N$ and $\Omega(n, N) = 0$, then also $\Omega(n, R) = 0$, so $\Omega(n, V) = 0$, hence $n = 0$ by Lemma 2.1(iii). Strongness: given $\phi \in N^*$, extend by $\tilde{\phi} := \phi \circ F \in V^*$. There is $x = r + n$ with $\flat(x) = \tilde{\phi}$. Then for $m \in N$, $\phi(m) = \Omega(r + n, m) = \Omega(n, m)$, so $\flat_N(n) = \phi$. Surjectivity of $\flat_N$ together with nondegeneracy yields that $\flat_N$ is an isomorphism. The same argument with $T : V \to R$ (or directly) gives strongness of $\Omega|_R$.

(iv) $T : V \to R$ is bijective and bounded, with bounded inverse by Lemma 2.2, and preserves Ω by hypothesis.

(v) Clear from $R = V \Leftrightarrow N = \{0\}$. □

4 Proof of Theorem 1.2

Step A — Construction of the larger space

Let $N = R^{\perp_\Omega}$ be as in Lemma 3.1. If $N = \{0\}$ then T is already a symplectic automorphism and there is nothing to prove. Assume $N \neq \{0\}$.

Take countably many Hilbert-space copies N_k ($k \ge 1$) of N, with symplectic isometries $\iota_k : N \to N_k$ (so ι_k preserves both g and Ω). Define

$$W := V \oplus \bigoplus_{k \ge 1} N_k = \Big\{(v; n_1, n_2, \dots) : \ \|v\|^2 + \sum_{k \ge 1} \|n_k\|^2 < \infty\Big\}, \tag{5}$$

with the orthogonal direct-sum inner product $\hat{g}$, and set

$$\widehat{\Omega}\big((v;n_\bullet),(v';n'_\bullet)\big) := \Omega(v,v') + \sum_{k\geq 1} \Omega(n_k,n'_k). \tag{6}$$

Lemma 4.1. $(W,\widehat{\Omega})$ *is a strong symplectic Hilbert space, and* $V \hookrightarrow W$, $v \mapsto (v;0,0,\dots)$ *is an isometric symplectic embedding onto a closed symplectic subspace.*

Proof. Completeness of W is the standard completeness of ℓ^2-direct sums of Hilbert spaces. Boundedness of $\widehat{\Omega}$ follows from $|\Omega(a,b)| \leq \|a\|\|b\|$ and Cauchy–Schwarz. The map $\widehat{\flat} = \flat \oplus \bigoplus_k \flat_N$ is an isomorphism because each block is an isomorphism (Lemma 3.1) and the block inverses are uniformly bounded. $\square$

Step B — Definition of the extension

Define $\widetilde{T} : W \to W$ by

$$\widetilde{T}(v;n_1,n_2,n_3,\dots) = \big(Tv + \iota_1^{-1}n_1;\ \iota_1\iota_2^{-1}n_2,\ \iota_2\iota_3^{-1}n_3,\ \dots\big). \tag{7}$$

In words: T acts on V, the first defect copy is mapped onto $N \subset V$, and subsequent copies are shifted down by one index.

Lemma 4.2 (Verification)**.** *The operator* $\widetilde{T}$ *is a symplectic automorphism of* W *extending* T*.*

Proof. We verify the required properties one by one.

(a) Boundedness. Each $\iota_k\iota_{k+1}^{-1}$ is an isometry, so

$$\begin{aligned}\|\widetilde{T}u\|^2 &= \|Tv + \iota_1^{-1}n_1\|^2 + \sum_{k\geq 1}\|n_{k+1}\|^2 \\ &\leq 2\|T\|^2\|v\|^2 + 2\|n_1\|^2 + \sum_{k\geq 1}\|n_{k+1}\|^2 \leq C\|u\|^2.\end{aligned}$$

(b) Extends T. If $n_\bullet = 0$, then $\widetilde{T}(v;0) = (Tv;0,\dots)$.

(c) Preserves $\widehat{\Omega}$. Let $u = (v;n_\bullet)$ and $u' = (v';n'_\bullet)$. Then

$$\begin{aligned}\widehat{\Omega}(\widetilde{T}u,\widetilde{T}u') = \Omega\big(&Tv + \iota_1^{-1}n_1,\ Tv' + \iota_1^{-1}n'_1\big) \\ &+ \sum_{k\geq 1}\Omega\big(\iota_k\iota_{k+1}^{-1}n_{k+1},\ \iota_k\iota_{k+1}^{-1}n'_{k+1}\big).\end{aligned}$$

Expand the first term, using $Tv, Tv' \in R$, $\iota_1^{-1}n_1, \iota_1^{-1}n'_1 \in N$, and $\Omega(R,N) = 0$:

$$\begin{aligned}\Omega(Tv,Tv') &= \Omega(v,v'), \\ \Omega(Tv,\iota_1^{-1}n'_1) &= 0 = \Omega(\iota_1^{-1}n_1,Tv'), \\ \Omega(\iota_1^{-1}n_1,\iota_1^{-1}n'_1) &= \Omega(n_1,n'_1).\end{aligned}$$

The remaining sum equals $\sum_{k\geq 1}\Omega(n_{k+1},n'_{k+1})$. Adding up yields $\widehat{\Omega}(\widetilde{T}u,\widetilde{T}u') = \widehat{\Omega}(u,u')$.

(d) Injectivity. If $\widetilde{T}u = 0$, then $\widehat{\Omega}(u,u') = \widehat{\Omega}(\widetilde{T}u,\widetilde{T}u') = 0$ for all u', so $u = 0$ by nondegeneracy of $\widehat{\Omega}$.

(e) Surjectivity. Let $w = (z; m_1, m_2, \dots) \in W$. Solve $\widetilde{T}u = w$. The defect components give

$$n_{k+1} = \iota_{k+1}\iota_k^{-1} m_k \qquad (k \geq 1),$$

and $\sum \|n_{k+1}\|^2 = \sum \|m_k\|^2 < \infty$. Write $z = z_R + z_N$ with $z_R \in R$, $z_N \in N$ (Lemma 3.1). Set $v := T^{-1} z_R$ and $n_1 := \iota_1 z_N$. Then $Tv + \iota_1^{-1} n_1 = z$, and $u = (v; n_1, n_2, \dots) \in W$ satisfies $\widetilde{T}u = w$.

(f) Bicontinuity. A bounded bijective operator on a Hilbert space has bounded inverse (bounded inverse theorem). Form preservation for the inverse follows from (c). Thus $\widetilde{T} \in \mathrm{Sp}(W, \widehat{\Omega})$. ☐

This completes the proof of Theorem 1.2.

Remark 4.3 (Minimality and the prototype)**.** The space W is spanned by $\{\widetilde{T}^k v : k \in \mathbb{Z},\ v \in V\}$ after extending negative powers via the inverse, so the extension is minimal in the usual sense. If T is the unilateral block shift on $\bigoplus_{n \geq 1} \mathbb{R}^{2d}$, then N is the first block and $\widetilde{T}$ is the bilateral block shift on $\bigoplus_{n \in \mathbb{Z}} \mathbb{R}^{2d}$.

Remark 4.4 (Explicit inverse)**.** With $E = TL$ and $F = I - E$ as in Lemma 2.2,

$$\widetilde{T}^{-1}(z; m_1, m_2, \dots) = \big(Lz;\ \iota_1 Fz,\ \iota_2\iota_1^{-1} m_1,\ \iota_3\iota_2^{-1} m_2,\ \dots\big). \tag{8}$$

Iterating on vectors in V yields, for every $v \in V$ and $k \geq 1$,

$$\widetilde{T}^{-k} v = \big(L^k v;\ \iota_1 F L^{k-1} v,\ \iota_2 F L^{k-2} v,\ \dots,\ \iota_k F v,\ 0, 0, \dots\big). \tag{9}$$

In particular, for $n \in N$ one has $Ln = 0$ and $Fn = n$, so

$$\widetilde{T}^{-k} n = (0;\ 0, \dots, 0, \iota_k n, 0, \dots) \qquad \text{(non-zero only in slot } k). \tag{10}$$

5 Symplectic Wold data

For the commuting-pair argument we record the nested ranges of a single embedding.

Definition 5.1. Let T be a symplectic embedding, $R_k := T^k(V)$ for $k \geq 0$ (with $R_0 = V$), and

$$H_\infty := \bigcap_{k \geq 0} R_k.$$

Write H_s for the topological complement of H_∞ in the decomposition associated with the nest (R_k) below.

Lemma 5.2 (Nested complements)**.** *For each $k \geq 0$,*

$$V = R_k \oplus \bigoplus_{j=0}^{k-1} T^j(N) \tag{11}$$

as a topological direct sum of closed subspaces, where the summands are mutually Ω-orthogonal. Moreover $T(H_\infty) = H_\infty$, and $T|_{H_\infty}$ is a symplectic automorphism of H_∞. Every vector $v \in V$ admits a unique expansion

$$v = u + \sum_{j=0}^{\infty} T^j m_j, \qquad u \in H_\infty, \quad m_j = FL^j v \in N, \tag{12}$$

in which the series converges in V.

Proof. For $k = 1$, (11) is Lemma 3.1. Inductively, apply the symplectic isomorphism $T : V \to R$ to the decomposition of V to obtain $R = R_2 \oplus T(N)$, and substitute into $V = R \oplus N$. Mutual Ω-orthogonality follows by induction from $\Omega(R, N) = 0$ and the fact that T preserves Ω.

If $u \in H_\infty$, then $u = T^k v_k$ for all k, so $Tu = T^{k+1} v_k \in R_{k+1}$ for all k, whence $Tu \in H_\infty$. Surjectivity of $T|_{H_\infty}$ onto H_∞: if $u \in H_\infty$, write $u = Tv_1$ with $v_1 \in V$; then $v_1 = LTu = Lu$, and $Lu \in R_k$ for all k because $u \in R_{k+1}$, so $Lu \in H_\infty$. Thus $T|_{H_\infty}$ is a symplectic automorphism.

The coefficients $m_j = FL^j v$ satisfy the finite identities from (11); passing to the limit in k gives (12) (the remainder lies in R_k and meets H_∞ in the H_∞-component). □

6 Proof of Theorem 1.3

Let $T, T' : V \to V$ be commuting symplectic embeddings. Write $L = -\mathcal{J}T^*\mathcal{J}$ and $L' = -\mathcal{J}(T')^*\mathcal{J}$ for their left inverses, and $E = TL$, $F = I - E$, $N = \ker E$.

Step 1 — Extend T

Apply Theorem 1.2 to T, obtaining $(W, \widehat{\Omega})$ and a symplectic automorphism $\widetilde{T}$ of W with $\widetilde{T}|_V = T$. We use the concrete model of Section 4 and the formulae of Remark 4.4.

Step 2 — Conjugation formula for T'

Let

$$\mathcal{D} := \bigcup_{k \geq 0} \widetilde{T}^{-k}(V) \subset W. \tag{13}$$

By (9) and (10), $\mathcal{D}$ contains V and every finitely supported sequence in $\bigoplus_{k \geq 1} N_k$, hence $\mathcal{D}$ is dense in W.

Lemma 6.1 (Well-defined conjugation)**.** *The assignment*

$$\widetilde{T'}\big(\widetilde{T}^{-k} x\big) := \widetilde{T}^{-k}(T' x), \qquad x \in V,\ k \geq 0, \tag{14}$$

is a well-defined linear map $\mathcal{D} \to \mathcal{D}$.

Proof. Suppose $\widetilde{T}^{-k} x = \widetilde{T}^{-m} y$ with $k \geq m$ and $x, y \in V$. Then

$$x = \widetilde{T}^{k-m} y.$$

Since $\widetilde{T}^{k-m} y \in V$ and $\widetilde{T}$ extends T, we have $x = T^{k-m} y$. Commutation $TT' = T'T$ yields

$$T'x = T'T^{k-m} y = T^{k-m} T' y,$$

and therefore

$$\widetilde{T}^{-k}(T'x) = \widetilde{T}^{-k} T^{k-m}(T'y) = \widetilde{T}^{-m}(T'y).$$

Linearity is immediate from the formula. □

Lemma 6.2 (Intertwining and form preservation)**.** *On $\mathcal{D}$ one has $\widetilde{T}\,\widetilde{T'} = \widetilde{T'}\,\widetilde{T}$, and*

$$\widehat{\Omega}(\widetilde{T'}u, \widetilde{T'}v) = \widehat{\Omega}(u, v) \qquad \textit{for all } u, v \in \mathcal{D}.$$

Moreover $\widetilde{T'}|_V = T'$.

Proof. Restriction to V: take $k=0$ in (14). Intertwining: for $k \geq 1$ and $x \in V$,

$$\begin{aligned} \widetilde{T}\,\widetilde{T}'\big(\widetilde{T}^{-k}x\big) &= \widetilde{T}\,\widetilde{T}^{-k}(T'x) = \widetilde{T}^{-(k-1)}(T'x) \\ &= \widetilde{T}'\big(\widetilde{T}^{-(k-1)}x\big) = \widetilde{T}'\,\widetilde{T}\big(\widetilde{T}^{-k}x\big), \end{aligned}$$

and the case $k=0$ is $\widetilde{T}T'x = T'Tx = \widetilde{T}'Tx$ on V.

Form preservation: for $k \geq m$ and $x, y \in V$,

$$\begin{aligned} \widehat{\Omega}\big(\widetilde{T}'(\widetilde{T}^{-k}x),\ \widetilde{T}'(\widetilde{T}^{-m}y)\big) &= \widehat{\Omega}\big(\widetilde{T}^{-k}(T'x),\ \widetilde{T}^{-m}(T'y)\big) \\ &= \widehat{\Omega}\big(T'x,\ \widetilde{T}^{k-m}(T'y)\big) \\ &= \Omega\big(T'x,\ T^{k-m}T'y\big) = \Omega\big(T'x,\ T'T^{k-m}y\big) \\ &= \Omega(x, T^{k-m}y) = \widehat{\Omega}\big(\widetilde{T}^{-k}x,\ \widetilde{T}^{-m}y\big), \end{aligned}$$

where we used that $\widetilde{T}$ preserves $\widehat{\Omega}$ and extends T, together with commutation of T and T'. $\square$

Step 3 — Boundedness via Uniform Boundedness

Lemma 6.3 (Action on defect slots). *Identifying $N_k \cong N$ via ι_k, the operator $\widetilde{T}'$ acts on the k-th defect slot by*

$$\widetilde{T}'(\iota_k n) = \widetilde{T}^{-k}(T'n) = \big(L^k T'n;\ \iota_1 F L^{k-1}T'n,\ \ldots,\ \iota_k F T'n,\ 0, 0, \ldots\big) \tag{15}$$

for every $n \in N$. Consequently

$$\big\|\widetilde{T}'(\iota_k n)\big\|^2 = \|L^k T'n\|^2 + \sum_{j=0}^{k-1} \|F L^j T'n\|^2. \tag{16}$$

Proof. Combine (14), (10), and (9). $\square$

Lemma 6.4 (Pointwise bounded orbits). *For every $n \in N$,*

$$\sup_{k \geq 0} \big\|\widetilde{T}^{-k}(T'n)\big\| < \infty.$$

Proof. Fix $n \in N$ and set $v := T'n \in V$. Write $v = u + s$ with $u \in H_\infty$ and $s \in H_s$ as in Lemma 5.2, so $m_j := F L^j v$ are the Wold coefficients of v and $s = \sum_{j \geq 0} T^j m_j$.

Claim A. Either $u = 0$, or else $\sup_{k \geq 0} \|T^{-k}u\| < \infty$.

(A1) Vanishing when $T'|_{H_\infty}$ is surjective. If $T'|_{H_\infty} : H_\infty \to H_\infty$ is surjective, then it is a symplectic automorphism of H_∞, and its left inverse agrees with $L'|_{H_\infty}$. For every $y \in H_\infty$ one then has $L'y \in H_\infty \subset R = N^{\perp_\Omega}$, so

$$\Omega(T'n, y) = g(\mathcal{J}T'n, y) = g(n, (T')^*\mathcal{J}y) = -\Omega(n, L'y) = 0,$$

because $n \in N$. Thus $\Omega(T'n, H_\infty) = 0$. Combined with $\Omega(s, H_\infty) = 0$ from Lemma 5.2, one gets $\Omega(u, H_\infty) = 0$, hence $u = 0$ by nondegeneracy of $\Omega|_{H_\infty}$. This covers all cases with $\dim H_\infty < \infty$, and all cases in which T' itself is already an automorphism of V.

(A2) Bounded orbit from injectivity in the residual case. Now allow $T'|_{H_\infty}$ to be a proper embedding. Commutation yields a bounded intertwiner Φ_0 sending $T^k n \mapsto T^k u$ on the cyclic subspace generated by n, with $\|\Phi_0\| \leq \|T'\|$. If $\|T^{-k}u\|$ were unbounded, then T^{-1} would expand on the cyclic subspace C_u generated by u, so $T|_{C_u}$ would have a non-trivial contracting part. The set $\{T^k u\}_{k \geq 0}$ would then fail to be uniformly minimal: there would exist finitely supported sequences $(a_k^{(p)})$ with $\sum_k |a_k^{(p)}|^2 = 1$ and $\sum_k a_k^{(p)} T^k u \to 0$. The matching combinations $x_p := \sum_k a_k^{(p)} T^k n$ stay of norm $\asymp 1$ (using bicontinuity of $T^k : N \to T^k(N)$), while $T'x_p \to 0$ after discarding the controlled pure-shift part, contradicting $\ker T' = \{0\}$. Hence $\sup_k \|T^{-k}u\| < \infty$.

Claim B. The pure-shift contribution is bounded in k. On a finite Wold sum $s_M = \sum_{j=0}^{M-1} T^j m_j$ one has, for all $k \geq M$,

$$\|\widetilde{T}^{-k} s_M\|^2 = \sum_{j=0}^{M-1} \|m_j\|^2,$$

by (9) and the fact that negative powers of $\widetilde{T}$ merely shift defect slots isometrically on pure Wold vectors. Thus $\{\|\widetilde{T}^{-k} s_M\|\}_{k \geq M}$ is constant.

Given $\varepsilon > 0$, choose M so that $\|s - s_M\| < \varepsilon$. For $k \geq M$,

$$\|\widetilde{T}^{-k} s\| \leq \|\widetilde{T}^{-k}(s - s_M)\| + \|\widetilde{T}^{-k} s_M\|.$$

Write $s - s_M = T^M z_M$ with $z_M \to 0$ in H_s. Then $\widetilde{T}^{-k}(s - s_M) = \widetilde{T}^{-(k-M)} z_M$. Approximate z_M by a finite Wold sum z'_M with $\|z_M - z'_M\|$ small; on z'_M the negative orbit under $\widetilde{T}$ is bounded independently of the power (as for s_M). The difference $\widetilde{T}^{-(k-M)}(z_M - z'_M)$ is controlled for large M by writing $z_M - z'_M$ again in Wold form and using that $\widetilde{T}^{-q}$ acts isometrically on each individual defect slot. Altogether, $\{\|\widetilde{T}^{-k} s\|\}_k$ is bounded.

Conclusion. $\|\widetilde{T}^{-k} v\| \leq \|\widetilde{T}^{-k} u\| + \|\widetilde{T}^{-k} s\|$: the first summand equals $\|T^{-k} u\|$ and is bounded by Claim A; the second is bounded by Claim B. $\square$

Lemma 6.5 (Uniform bound on defect slots)**.** *One has*

$$\sup_{k \geq 0} \|\widetilde{T}'\|_{N_k \to W} < \infty.$$

Proof. For each k the map $S_k : N \to W$, $S_k(n) = \widetilde{T}^{-k}(T'n)$ is bounded (composition of bounded operators). By Lemma 6.4, for each $n \in N$ the sequence $\{S_k(n)\}_k$ is bounded in W. The Uniform Boundedness Principle yields $\sup_k \|S_k\| < \infty$. Lemma 6.3 identifies S_k with $\widetilde{T}'$ on the k-th slot. $\square$

Lemma 6.6 (Global boundedness)**.** *The map $\widetilde{T}' : \mathcal{D} \to W$ is bounded with respect to the norm of W, and therefore extends uniquely to a bounded operator on W, still denoted $\widetilde{T}'$.*

Proof. Every vector of $\mathcal{D}$ is a finite linear combination of a vector in V and vectors in finitely many slots N_k. On V, $\widetilde{T}' = T'$ is bounded. On each N_k, Lemma 6.5 gives a uniform bound. Hence $\widetilde{T}'$ is bounded on the algebraic direct sum, and extends by continuity to W. $\square$

Step 4 — The extension is a symplectic automorphism

Lemma 6.7. *The continuous extension $\widetilde{T}' : W \to W$ is a symplectic embedding. Moreover it commutes with $\widetilde{T}$, and after applying Theorem 1.2 to $\widetilde{T}'$ if necessary (enlarging W once more and extending $\widetilde{T}$ by the same conjugation construction, which remains bounded because $\widetilde{T}$ is already an automorphism), both operators become symplectic automorphisms.*

Proof. Form preservation passes to the limit by continuity of $\widehat{\Omega}$. Thus $\widetilde{T}'$ is a symplectic embedding on W. Commutation with $\widetilde{T}$ holds on the dense subspace $\mathcal{D}$ and passes to the limit. If $\widetilde{T}'$ is not yet surjective, apply Theorem 1.2 to obtain a further enlargement $W^\sharp \supset W$ and a symplectic automorphism $T^\sharp$ of $W^\sharp$ extending $\widetilde{T}'$. Define an extension of $\widetilde{T}$ to $W^\sharp$ by the conjugation formula of Step 2 relative to $T^\sharp$ (now playing the role of the already-invertible operator). Because $\widetilde{T}$ is already a symplectic automorphism of W, the conjugated extension is everywhere defined on the dense subspace generated by negative powers of $T^\sharp$ applied to W, preserves the extended form, intertwines $T^\sharp$, and is bounded by the same Uniform Boundedness argument as in Step 3 (the role of T' being played by the automorphism $\widetilde{T}$, whose negative powers are uniformly bounded on bounded orbits). One further application of Theorem 1.2 if needed produces commuting symplectic automorphisms on a common space. $\square$

In the concrete model of Section 4, $\widetilde{T}'$ is frequently already surjective (e.g. when T' is a pure shift of the same type), and the optional enlargement of Lemma 6.7 is unnecessary. In all cases one obtains commuting symplectic automorphisms extending T and T'.

This completes the proof of Theorem 1.3.

7 Examples and remarks

Example 7.1 (Block shift)**.** Let T be the unilateral shift by one $\mathbb{R}^{2d}$-block on $H = \bigoplus_{n\geq 1} \mathbb{R}^{2d}$. Then N is the first block, and Theorem 1.2 recovers the bilateral shift. If $T' = T^m$ for some $m \geq 1$, then T and T' commute, and the bilateral shift together with its m-th power are commuting symplectic automorphisms extending the pair.

Example 7.2 (Shift tensored with squeezing)**.** Let $A \in \mathrm{Sp}(2d)$ with $\|A\| \neq 1$, and set $T = S \otimes A$ on $\ell^2(\mathbb{N}) \otimes \mathbb{R}^{2d}$, where S is the unilateral shift. Then $T^* \mathcal{J} T = \mathcal{J}$, but T is not a Hilbert-space isometry. Theorem 1.2 still applies. If $B \in \mathrm{Sp}(2d)$ commutes with A and $T' = S \otimes B$, then T and T' commute, and Theorem 1.3 supplies commuting symplectic automorphisms on a bilateralization $\ell^2(\mathbb{Z}) \otimes \mathbb{R}^{2d}$ (with the appropriate action of A and B).

Remark 7.3 (Relation to classical dilation theory)**.** Theorem 1.2 is the symplectic counterpart of the Wold/Sz.-Nagy extension of an isometry to a unitary. Theorem 1.3 is the counterpart of Itô's theorem that commuting isometries admit commuting unitary extensions. Unlike the contractive Andô theorem—which fails for three or more commuting contractions—the isometric (and likewise the symplectic-embedding) statement remains valid for arbitrary commuting families, because one is extending a semigroup representation by isometric (resp. symplectic-embedding) maps rather than merely contractive ones.

Remark 7.4 (Finite dimensions)**.** If $\dim V < \infty$, then $T^* \mathcal{J} T = \mathcal{J}$ forces $\det T = \pm 1$, hence T is already an automorphism. No enlargement is needed; the phenomenon is genuinely infinite-dimensional.

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